

Sponges

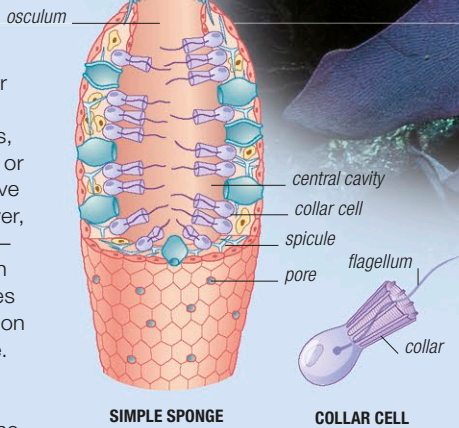
PHYLUM	Porifera
CLASSES	3
ORDERS	24
FAMILIES	127
SPECIES	About 10,000

Sponges, which were once thought to be plants, are believed to be Earth's first animals. Unlike most animals, which move through their habitat, sponges are immobile, living attached to solid surfaces and feeding by setting up

currents of water from which they filter out particles of food. In other words, their aquatic environment is made to move through their body rather than the other way around. Most sponges are marine animals, although some species have adapted to life in freshwater.

Anatomy

The body of a sponge lacks many features typical of other animals. Although the cells perform specialized functions, they do not form true tissues or organs, and there are no nerve and few muscle cells. However, sponges do have a skeleton—a network of organic collagen fibers and inorganic structures (called spicules) made of silicon dioxide or calcium carbonate. Sponges have no system of symmetry, nor do they have clearly distinct body parts. The simplest sponge is essentially a tube, closed at one end, but complex sponges take on many forms, from crusts, spheres, and barrels to tangled branches.



SIMPLE SPONGE

COLLAR CELL

BODY SECTION

Water enters a simple sponge through minute pores in the outer surface. It then passes into a central cavity before being expelled through a larger aperture (the osculum).

Feeding

Sponges are remarkably effective filter feeders. The cells lining the body's inner surface are known as collar cells, each consisting of a ring (or collar) of tentacle-like structures surrounding a whiplike feature called the flagellum. Movement of the flagella creates the current of water that moves through the body. As the water passes through, plankton and particles of organic matter are trapped in the collars.



CARNIVOROUS SPONGE

Found on deep-sea bottoms and in Mediterranean caves, this unusual sponge catches live prey on hooklike spicules at the end of long filaments and digests it externally.

Class Calcarea

Calcareous sponges

Occurrence c.650 spp. worldwide; in shallow water (in crevices, overhangs and caves, on reefs and on seaweed)



These are the only sponges that have skeletal spicules made of calcium carbonate, the mineral that mollusks use to make their shells. Compared to other sponges, they are mostly small—less than 4 in (10 cm) high—lobe- or tube-shaped and drab, but some reef species are brightly colored.



LEUCETTUSA LANCIFER
This flask sponge, less than 2 in (5 cm) high, has the classic compound vase shape of calcareous species. Members of the genus are widespread in shallow waters from the tropics to the subantarctic.

Class Hexactinellida

Glass sponges

Occurrence c.500 spp. worldwide in deep cold water; mostly at depths of 650–6,000 ft (200–2,000 m) in seas



Most glass sponges are less than 3¼ ft (1 m) high but in some areas many grow together to form “sponge reefs” up to 65 ft (20 m) in height. Their siliceous spicules are usually fused together to form rigid, often beautiful, glasslike masses. These structures are fragile, however, and the skeletons of the species living in deep water tend to collapse when they are dredged up and brought to the surface.

VENUS' FLOWER BASKET

Glass sponges are often objects of remarkable beauty. Species in the genus Euplectella, such as the one shown here, are found in tropical oceans below a depth of about 500 ft (150 m). They grow up to 12 in (30 cm) tall.



Class Demospongiae

Demosponges

Occurrence c.7,200 spp. worldwide; in shallow and deep water, on solid surfaces of all kinds



These sponges make up nearly 90 percent of all sponge species, and include the familiar bath sponge, whose populations are now badly affected by overcollection. Demosponges vary in size from less than ¼ in (1 cm) to well over 3¼ ft (1 m) in height or breadth, and are often brilliantly colored, creating underwater “gardens” that rival corals for their vivid hues. While crusts and mounds are common, these sponges come in a variety of shapes—some even bore into mollusk shells, leaving them riddled with pin-sized holes.

STOVE-PIPE SPONGE

Aplysina archeri, or stove-pipe sponge, can reach a length of over 20 in (50 cm). In sponges with this kind of shape, water is drawn in through pores in the sides of the tube, and expelled through an opening at the top.



CARIBBEAN PINK VASE SPONGE

Many demosponges form irregular masses, but many others are funnel- and vase-shaped like Niphates digitalis (above). Up to 12 in (30 cm) tall, it has, like most demosponges, a skeleton of siliceous spicules and collagen fibers.

